MIL-R-21530/20A (WEPS)
26 February 1962
Superseding
MIL-R-21530/20 (WEPS)
29 August 1961

MILITARY SPECIFICATION
RESOLVERS, ELECTRICAL, AC
FOR USE IN
POLARIS FLEET BALLISTIC MISSILE WEAPON SYSTEM
TYPE 41Ru47N8a

1. SCOPE

1.1 This specification establishes the requirements for a Resolver, Electrical, AC, Type 41RU47N8a, hereinafter termed resolver.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-21530	Resolver, Electrical, AC, General Specification for

DRAWINGS

Bureau of Naval Weapons

2259514	Resolver, Type 41RU47N8a

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The resolver shall be in accordance with MIL-R-21530 with the following exceptions:

FSC-5990

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(a) Resolver shall be tested as an RC only.

(b) Resolver shall be designed to operate under conditions of 3000 pounds, compressive, axial load across the rotor.

3.2 Design and construction. - The resolver shall be of the design, construction, and physical dimensions as specified on Dwg. 2259514.

3.3. Specific characteristic requirements. - The resolver shall have specific characteristic requirements as follows:

Input voltage	115 volts root mean square (v rms) ± 10 percent
Test voltage	115 v rms ± 2 percent
Nominal input frequency	800 cycles per second (cps)
Test frequency	800 cps ± 0.5 percent
Primary	Stator
Temperature rise	Five degrees centigrade (C) max. at 25 degrees C
Friction torques max	200 gram-centimeters at room temp 300 gram-centimeters at -55 and +85 degrees C
Rotor impedance, Zro	See note a
Stator impedance, Zso	See note a
Rotor impedance, Zrs	See note a
Stator impedance, Zss	See note a
Total null voltage, max	50 millivolts (mv) at room temp 75 mv at minus 55 and plus 85 degrees C
Angular error versus voltage	5 seconds max.
Angular error versus frequency	5 seconds max.
Angular error versus vibration	10 seconds max.
Angular error versus humidity	5 seconds max.
Transformation ratio K_e	See note b
Phase shift, stator to rotor	Zero degrees ± 0.6 degrees
Angular error	20 seconds max at room temp.
Null shift/para. 3.5.8 of MIL-R-21530	20 seconds max. at -55 and +85 degrees C

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Angular interval	5 degrees
Impedance balance	15 Z-seconds max.
Primary current, open circuit	0.0242 amperes, nominal
Primary power, open circuit	1.11 watts, nominal
Capacity R_1-R_2	4000 pf max.

Note a: Requirement shall be as specified on Dwg. 2259414.

Note b: Resolver is externally trimmed to achieve 0.1 percent accuracy of transformation ratio. Accuracy is achieved by trimming resistors as specified on Dwg. 2259414. The resolver must meet specified impedances with trimming resistors in place.

4. QUALITY ASSURANCE PROVISIONS

4.1 The sampling and inspection shall be in accordance with MIL-R-21530.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-21530.

6. NOTES

6.1 The notes of MIL-R-21530 are applicable herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-21530/20A(WP) RESOLVERS, ELECTRICAL AC TYPE 41Ru47N8a

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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DEPARTMENT OF THE NAVY
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